Project Proposal: Students Ergonomics Monitoring

Core Concept:
A camera-based health monitor and simple wearable devices that track heart rate, posture and blinking while students work at their laptop. The sensors will provide a notification when notice fatigue, stress or bad posture is measured.

How it connects to our syllabus:

1. We will measure the openness of the eyes using the pc webcam and a ML model. We measure the heartbeat with the wearable (optionally also the temperature through the patch), and we will implement the human pose estimation (nearable) to check if your shoulders and neck are slouching.

2. The Processing: we will apply the signal processing and Fourier Transform techniques to clean up the noisy heart rate signals if necessary. We can use a lightweight machine learning model to simply classify your current posture as "Good" or "Slouching". Finally, we will implement the ML model for measuring the fatigue level through the eyes blinking rate.

3. The Dashboard (streamlit): we will build a simple GUI where you can click "Start Monitoring" and see a real-time graph of your vitals and it will show the outcome of the measurements. Finally, it will show some statistic of real working students.

4. **OPTIONAL:** The OpenClaw Integration: Our ui app will quietly save our vitals (e.g., average heart rate, minutes slouched) to a local CSV file every 10 minutes. Our will give our OpenClaw agent a simple custom "skill" (a markdown instruction file) that tells it to read that CSV file periodically. If OpenClaw sees your heart rate is elevated and you've been slouching, it will use its native integrations to text your phone: "Hey, you've been slouching for 20 minutes and your resting heart rate is creeping up. Time to stand up and drink some water!"